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23 July 1984
SUPERSEDING
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MILITARY SPECIFICATION

REFILL KIT, ANALYZING COMPONENTS, CBR AGENT SAMPLING AND ANALYZING KIT, M33

This specification is approved for use by US Army Armament, Munitions and Chemical Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a refill kit of miscellaneous analyzing components.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

SS-P-166	-	Pencils, Lead
SS-P-196	-	Pencils: Spiral-Paper-Form; Wax, Glass and China Marking
UU-P-21	-	Pads, Writing Paper
UU-P-258	-	Paper; Index
ZZ-S-751	-	Stoppers, Bottle, Rubber

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, Armament Research and Development Center, US Army Armament, Munitions and Chemical Command, ATTN: DRSMC-TSC-S(A), Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6665

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PPP-B-566 - Boxes, Folding, Paperboard
PPP-B-576 - Box, Wood, Cleated, Veneer, Paper Overlaid
PPP-B-585 - Boxes, Wood, Wirebound
PPP-B-636 - Box, Shipping, Fiberboard
PPP-F-320 - Fiberboard: Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes

MILITARY

MIL-P-116 - Preservation, Methods of
MIL-D-51082 - Detector Tubes, Chemical Agent
MIL-D-51083 - Detector Ticket, Anticholinesterase Agent
MIL-S-51189 - Substrate Solution and Dispenser
MIL-R-51197 - Reagents, Solvents, and Test Papers for Sampling and Analyzing Kit, CBR Agents
MIL-P-51409 - Paper, Chemical Agent Detector, VGH, ABC-M8; Booklet of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

DRAWINGS

US ARMY ARMAMENT, MUNITIONS AND CHEMICAL COMMAND

CHEMICAL RESEARCH AND DEVELOPMENT CENTER

5-67-30 - Container, Swabs
5-77-449 - Seals, End, Detector Tubes
5-77-451 - Box, Test Tubes
5-77-482 - Envelope Shipping Tag
5-77-701 - Matches, Packaging
5-77-720 - Pad, Field, Notebook
5-77-726 - Tubes, Capillary
5-77-911 - Dispenser, Filter Paper
5-77-951 - Clip, Wood
5-77-2349 - Marking for Shipping Container Refill Kit - M33

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

ASTM STANDARDS

D 1193 - Reagent Water
D 3951 - Commercial Packaging

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Materials and components. All materials and components used in the assembly of the kit shall conform to the specifications and drawings forming a part of this specification.

3.2 Contents. The refill kit shall be composed of components listed in table I in the quantities specified.

TABLE I. Contents

Item	Component	Quantity	Applicable document
1	Paper, Chemical Agent Detector, VGH, ABC-M8	2 booklets	MIL-P-51409
2	Detector tubes (51 clips; see Dwg 5-77-387)	1 set	MIL-D-51082
3	Alcohol tubes	2 boxes	MIL-R-51197
4	Benzene tubes	1 box	MIL-R-51197
5	Flannel swabs	4 packs	Dwg 5-67-30
6	Capillary tubes	1 vial	Dwg 5-77-726
7	Combination filters	1 box	MIL-R-51197
8	Reagent containers	2 sets	MIL-R-51197
9	Detector tube end seals	1 box	Dwg 5-77-449
10	Safety matches	1 box	Dwg 5-77-701
11	Dragendorff papers	1 box	MIL-R-51197

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TABLE I. Contents (cont'd)

Item	Component	Quantity	Applicable document
12	Field notebook	2 pads	Dwg 5-77-720
13	Test tubes, micro	4 boxes	Dwg 5-77-451
14	Detector tickets	240	MIL-D-51083
16	Substrate dispenser (2 per box)	1 box	MIL-S-51189
17	Ball point pen, medium point, blue-black ink, w/cap and clip	1 ea	Commercial product
18	Lead pencil	1 ea	SS-P-166, type III
19	Red grease pencil (both ends dipped in collodium, 1/2 inch min)	1 ea	SS-P-196, class B
20	Return envelopes	15 ea	Dwg 5-77-482
21	Filter paper dispenser	1 ea	Dwg 5-77-911
22	Instruction card	1 ea	see 3.2.1
23	Wicks for alcohol burner, braided, cotton cloth, 1/4 in. diameter by 6 in. long	3 ea	Commercial product
24	Light bulb	2 ea	General Electric Co. PR18
25	Memorandum pad, 3 by 5 inches	1 ea	UU-P-21
26	Rubber stoppers, size 00	2 ea	ZZ-S-751, type A, grade 1
28	Plastic knitting needle, 4 in. long, 1/8 in. diameter	1 ea	Commercial product
29	Forceps, specimen jar, 175 mm long	1 ea	Fisher Scientific Co. Cat. No. 10-316
30	Forceps, straight, 4-1/2 inch long, teflon coated tip, min 1-1/2 inch, medium point	1 ea	Commercial product
31	Wood clips	1 set	Dwg 5-77-951
32	Bottle, screw cap, polyethylene, capacity 4 oz. (NSN 6640-00-229-8961). Each bottle shall be filled with 100 mL of distilled water* and a 0.010 in. thick polyethylene sheet heat sealed across the mouth of the bottle to prevent leakage.	2	Fisher Scientific Co. Pittsburgh, PA., Cat. No. 2-923
33	Needles, disposable, size D(18g), length 1-1/2 in.	2	Fisher Scientific Co. Pittsburgh, PA., Cat. No. 14-826-5
34	Syringe, plastipak, disposable 2.5 cc capacity with 1/10 cc subdivisions**	1	Fisher Scientific Co. Pittsburgh, PA., Cat. No. 14-823-50

*Distilled water shall conform to ASTM D 1193.
**The scale shall be covered with a transparent, pressure-sensitive, adhesive tape.

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3.2.1 Instruction card.

3.2.1.1 Material. The instruction card shall be made of white paper, approximately 3 inches by 4 inches by 0.0105 inch, conforming to grade C of UU-P-258.

3.2.1.2 Printing. The following instructions shall be printed in sans serif type on the instruction card with black printer's ink:

INSTRUCTIONS

1. Remove all depleted boxes of expendable components from the M19 kit and discard.
2. Place the refill components into the correct spaces within the M19 kit. Remove the box tops on reagent containers and Dragendorff paper. Remove the combination filters from the box only when resupplying the aluminum case.

3.3 Workmanship. Workmanship shall be such that the refill kit shall be free from damage (breakage, tears, or cracks) or foreign matter (grease, oil, or viscous material) that could affect the serviceability of the components.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the refill kits assembled by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions. However, no more than one lot of each type of detector tubes and detector tickets or one lot of combination filters, Dragendorff papers, and chemical agent detection paper, shall be represented in one lot of refill kits.

4.2.2 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105.

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4.2.3 Inspection procedure for examination. The sample unit for examination of packaging shall be one unit, intermediate, or shipping container as applicable. Sample unit, intermediate, and shipping containers and the sample refill kits shall be examined in accordance with the following classification of defects:

(a) Refill Kit, Analyzing Components, CBR Agents Sampling and Analyzing Kit, M33.

Categories		Defects
Critical:		None defined
Major:	AQL 1.0 percent defective	
	101	Component missing
	102	Component incorrectly assembled
	103	Marking illegible
	104	Marking missing or incorrect
	105	Workmanship (3.3)

(b) Packaging (see section 5).

Categories		Defects
Critical:		None defined
Major:	AQL 1.0 percent defective	
	101	Contents per container not as specified
	102	Container not as specified
	103	Container closure not as specified
	104	Pads missing or not as specified (when required)
	105	Container damaged
	106	Marking incorrect, missing, or illegible

5. PACKAGING

5.1 Unit packing. Unit packing shall be level A or commercial as specified (see 6.2).

5.1.1 Level A. Items in the quantity listed in table I shall be unit packed level A as specified in table II.

5.1.1.1 Intermediate pack. Items unit packed as specified in 5.1.1 shall be intermediately packed together in a fiberboard box conforming to grade W5c of PPP-B-636. The inside dimensions of the box shall be 24 by 8-1/4 by 7-1/2 inches. The box shall be closed by taping as specified in PPP-B-636 for exterior containers.

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TABLE II. Unit pack

Item No.	MIL-P-116 method
1	IC-1 or IC-3
2	IC-2*
3	IC-1 or IC-3
4	IC-1 or IC-3
5	IC-2*
6	IC-2*
7	IC-1 or IC-3
8	IC-1 or IC-3
9	IC-1 or IC-3
10	IC-1 or IC-3
11	IC-1 or IC-3
12	IC-1 or IC-3
13	IC-1 or IC-3
14	IC-2*
16	IC-2*
17, 18, 19	Unit packed together, III
20	IC-1 or IC-3
21	IA-8
22	Not applicable
23	IC-1 or IC-3
24	III
25	IC-1 or IC-3
26	III
28	III
29, 30	Unit packed together, IA-8
31	III
32	IA-8
33	IC-3
34	IC-3

*Unit box conforming to PPP-B-566

5.1.2 Commercial. The components of the refill kit shall be unit packed together commercially in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A or commercial as specified (see 6.2).

5.2.1 Level A. Four refill kits, intermediately packed as specified in 5.1.1.1, shall be packed in a shipping container having inside dimensions of 35-3/4 by 24-7/8 by 8-3/8 inches. The container shall conform to class 3, style optional of PPP-B-585. Each box shall be for a type 2 average load. Pads formed from material conforming to grade V3c of PPP-F-320 shall be inserted in the box as needed to prevent motion of contents. Each box shall be closed and reinforced as specified in the appendix to PPP-B-585.

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5.2.2 Commercial. Refill kits shall be packed commercially in accordance with ASTM D 3951.

5.3 Marking. Level A unit and intermediate containers shall be marked in accordance with MIL-STD-129. Level A shipping containers shall be marked as shown on Drawing 5-77-2349. Commercial unit packs and packs shall be marked in accordance with ASTM D 3951.

5.3.1 Special marking. The discard date, where applicable, and lot number shall be shown for each item on the unit container. The earliest discard date for any item in the intermediate pack or shipping container shall be shown immediately beneath the identification marking as follows: DISCARD _____ (Insert month and year of discard, as for example: 9/80).

6. NOTES

6.1 Intended use. The refill kit covered by this specification is intended to replace analyzing components of the Sampling and Analyzing Kit, CBR Agents, M19.

6.2 Ordering data. Acquisition documents should specify the title, number, and date of this specification, and the level of unit packing and packing required (see 5.1 and 5.2).

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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